



Modelling Financial Markets

Seminar 4: Portfolio Optimization — Hands-on



Daniel Traian PELE
Antoaneta Amza

Bucharest University of Economic Studies
IDA Institute Digital Assets
Blockchain Research Center
AI4EFin Artificial Intelligence for Energy Finance
Romanian Academy, Institute for Economic Forecasting
MSCA Digital Finance

Acronyms used in this material

- ▣ AI: Artificial Intelligence
- ▣ BET: Bucharest Exchange Trading (index)
- ▣ BET-TR: BET Total Return (index)
- ▣ BNR: Banca Națională a României (the National Bank of Romania)
- ▣ BVB: Bursa de Valori București (the Bucharest Stock Exchange)
- ▣ ERC: Equal Risk Contribution portfolio
- ▣ GMV: Global Minimum-Variance portfolio
- ▣ GMV-LW: Global Minimum-Variance portfolio with Ledoit–Wolf covariance
- ▣ HAC: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- ▣ HRP: Hierarchical Risk Parity
- ▣ IEF: iShares 7–10 Year Treasury Bond ETF
- ▣ MV: Mean–Variance tangency portfolio estimated from sample moments
- ▣ MV-LO: Mean–Variance tangency portfolio, Long-Only
- ▣ OOS: Out-Of-Sample
- ▣ SR: Sharpe Ratio
- ▣ TO: TurnOver
- ▣ US: United States

Outline

Introduction

Key Definitions

Part A: Pen-and-Paper Exercises

Part B: Backtests and Inference

Part C: Open Question

Wrap-up

References

Seminar Objectives

By the end of this seminar you can

- ▣ Derive GMV, ERC and Black–Litterman portfolios in closed form
- ▣ Run an out-of-sample backtest with a rolling window, turnover and costs
- ▣ Test a difference in Sharpe ratios with HAC errors and a block bootstrap
- ▣ Judge whether an optimised portfolio really beats 1/N

- ▣ Background: Lecture 4; textbook: [Franke, Härdle & Hafner \(2019\)](#), Part I
- ▣ Seminar: Drd. Antoaneta Amza, antoaneta.amza@csie.ase.ro

Structure

- ▣ **Part A:** eight pen-and-paper exercises (A1–A8)
- ▣ **Part B:** eight computational exercises, US and Bitcoin data (B1–B8)
- ▣ **Part C:** one open question on BVB portfolios, a starting point for the team project (C1), and a critique of an AI answer (C2)

How to Work in This Seminar

- ▣ **[Solved]** problems show the full worked solution; **[Proposed]** problems show only the task: their solutions are discussed in class
- ▣ Every Part B exercise ends with an **interpretation question**
 - ▶ It cannot be answered by reading a number: ask what would change the conclusion
- ▣ Out of sample means: the weights of month t use only data up to month $t - 1$
- ▣ Report every Sharpe ratio difference with a p -value or an interval

Solved and Proposed Problems

Solved: worked examples in this deck

- Part A: A1 (two-asset GMV), A3 (ERC with two assets), A5 (Black–Litterman, one view), A7 (Sharpe test under i.i.d. returns)
- Part B: B1 (1/N versus Markowitz, sectors), B2 (shrinkage and GMV risk), B3 (ERC and HRP, sectors)

Proposed: you solve them

- Part A: A2 (model: A1); A4 (model: A3); A6 (model: A5); A8 (model: A7)
- Part B: B4 (model: B2); B5 (model: B3); B6, B8 (model: B1); B7 (model: A5, A6)
- Part C: C1, open question (model: B1, B3); C2, critique of an AI answer (model: A7, B1)

Setup: Python Environment and Data

-  **Open in Colab:** `chapter4_seminar_notebook.ipynb`
- Google Colab (recommended) or local Python ≥ 3.10 with `numpy`, `pandas`, `scipy`, `matplotlib`
- Series: US sector and multi-asset ETFs, Bitcoin, eight BVB blue chips, BET-TR (course repository, `data/market`)
- One-month risk-free rate: [Kenneth French Data Library](#); USD/RON reference rate: [BNR](#)
- ETFs and stocks: dividend-adjusted prices; indices and Bitcoin: closing values
- Joint analyses: join the *prices* on common days first, then take month-end returns

Key Definitions: Portfolios

- Mean–variance (Markowitz (1952)): tangency $w_{MV} \propto \Sigma^{-1}\mu$; GMV $w_{GMV} \propto \Sigma^{-1}1$; both normalised to sum to one
- Two assets: $w^* = (\sigma_2^2 - \rho\sigma_1\sigma_2)/(\sigma_1^2 + \sigma_2^2 - 2\rho\sigma_1\sigma_2)$
- Risk contribution $RC_i = w_i(\Sigma w)_i/\sigma_p$; ERC (Maillard, Roncalli & Teïletche (2010)): equal RC_i
- HRP (López de Prado (2016)): correlation tree, then recursive bisection with inverse-variance weights
- Ledoit–Wolf (Ledoit & Wolf (2004)): $\hat{\Sigma} = \delta F + (1 - \delta)S$, constant-correlation target F
- Black–Litterman (Black & Litterman (1992)): prior $\pi = \delta_{RA}\Sigma w_b$, posterior μ_{BL} = precision-weighted mean of π and the views

Key Definitions: Evaluation and Inference

- ▣ Sharpe ratio $SR = \bar{r}^e / s(r^e)$, annualised $\times \sqrt{12}$; turnover $TO_t = \sum_i |w_{i,t} - w_{i,t-1}^+|$, $w^+ =$ drifted weights
- ▣ Net return: $r_t^{\text{net}} = r_t - c TO_t$, c in basis points (1 bp = 0.01%)
- ▣ i.i.d. test (Jobson & Korkie (1981); formula in Ledoit & Wolf (2008)), monthly SR_1, SR_2 , correlation ρ

$$\text{Var}(\widehat{SR}_1 - \widehat{SR}_2) \approx \frac{1}{T} \left[2 - 2\rho + \frac{1}{2} (SR_1^2 + SR_2^2 - 2SR_1SR_2\rho^2) \right]$$

- ▣ HAC test (Ledoit & Wolf (2008)): delta method with a Newey–West covariance of (r_1, r_2, r_1^2, r_2^2)
- ▣ Circular block bootstrap (Künsch (1989)): blocks of 6 months, pairs resampled together

Exercise A1: Two-Asset GMV, Stocks and Treasuries [Solved]

Task

- ▣ SPY and TLT, monthly 2002–2026: $\sigma_1 = 14.72\%$, $\sigma_2 = 13.49\%$, $\rho = -0.100$; means 11.73% and 4.32%
- ▣ Derive the minimum-variance weight from $\partial\sigma_p^2/\partial w = 0$
- ▣ Compute w^* , the volatility and the mean of the mix

Solution

- ▣ $\partial\sigma_p^2/\partial w = 2w\sigma_1^2 - 2(1-w)\sigma_2^2 + 2(1-2w)\rho\sigma_1\sigma_2 = 0$
- ▣ $w^* = \frac{0.01820+0.00199}{0.02166+0.01820+0.00399} = 0.461$
- ▣ $\sigma_p = 9.43\%$; mean
 $0.461(11.73\%) + 0.539(4.32\%) = 7.73\%$
- ▣ Volatility is 36% below SPY's with a negative correlation

Exercise A2: Two-Asset GMV, Stocks and Gold [Proposed]

Solution discussed in class

Task

- ▣ SPY and GLD, monthly December 2004 – July 2026: $\sigma_1 = 14.83\%$, $\sigma_2 = 17.18\%$, $\rho = 0.083$
- ▣ Compute w^* in SPY and the volatility of the mix
- ▣ What would the volatility of the same mix be if $\rho = 1$?
- ▣ Model: A1

Exercise A3: ERC with Two Assets [Solved]

Task

- Show that for two assets the ERC weights are $w_i \propto 1/\sigma_i$, for any ρ
- Apply to SPY ($\sigma_1 = 14.72\%$) and IEF ($\sigma_2 = 6.64\%$), $\rho = -0.100$, monthly 2002–2026

Solution

- $RC_1\sigma_p = w_1^2\sigma_1^2 + w_1w_2\rho\sigma_1\sigma_2$;
 $RC_2\sigma_p = w_2^2\sigma_2^2 + w_1w_2\rho\sigma_1\sigma_2$
- The cross terms are equal, so
 $RC_1 = RC_2 \iff w_1\sigma_1 = w_2\sigma_2$
- $w_1 = \frac{1/0.1472}{1/0.1472+1/0.0664} = 0.311$ in SPY
- Numerical ERC solver: 0.311; risk shares 50%/50%

Exercise A4: Who Carries the Risk in 60/40? [Proposed]

Solution discussed in class

Task

- ▣ Same SPY and IEF inputs as A3; covariance $\sigma_{12} = -0.00098$
- ▣ Compute the risk shares of a 60% SPY, 40% IEF portfolio and its volatility
- ▣ Compare with the ERC volatility
- ▣ Model: A3

Exercise A5: Black–Litterman with One View [Solved]

Task

- One asset: implied return $\pi = 5\%$, volatility $\sigma = 16\%$, $\tau = 0.05$
- View: expected return $q = 8\%$, with standard deviation 2%
- Compute the posterior mean and its standard deviation

Solution

- Prior variance $\tau\sigma^2 = 0.00128$ (precision 781); view variance 0.0004 (precision 2500)
- $\mu_{BL} = \frac{781(5\%) + 2500(8\%)}{3281} = 7.29\%$
- Weight of the view: $2500/3281 = 0.76$
- Posterior standard deviation: $1/\sqrt{3281} = 1.75\%$

Exercise A6: The Sector View of the Lecture [Proposed]

Solution discussed in class

Task

- ▣ Nine sectors, benchmark $1/N$, $\delta_{RA} = 2.5$, $\tau = 0.05$; implied XLK minus XLU: 2.22%
- ▣ View: XLK minus XLU = 3%, $\Omega = \tau P \Sigma P^T$
- ▣ Show that the posterior view is the simple average of prior and view; compute it
- ▣ Model: A5

Exercise A7: Is GMV Better than 1/N? [Solved]

Task

- ▣ Sector backtest, $T = 271$ months: monthly Sharpe GMV 0.2051, 1/N 0.1908, correlation 0.694
- ▣ Test H_0 : equal Sharpe ratios with the i.i.d. formula

Solution

- ▣ Bracket:
 $2 - 1.388 + 0.5(0.0421 + 0.0364 - 0.0377) = 0.6324$
- ▣ Standard error $\sqrt{0.6324/271} = 0.0483$
- ▣ $z = 0.0143/0.0483 = 0.30$, $p = 0.77$ (HAC: $p = 0.75$)
- ▣ Annualised: 0.71 vs 0.66, a difference far from significant

Exercise A8: ERC versus 1/N [Proposed]

Solution discussed in class

Task

- ▣ Same backtest: monthly Sharpe ERC 0.2018, 1/N 0.1908, correlation 0.9927
- ▣ Compute z and p ; compare with A7
- ▣ Why can a smaller difference be closer to significance?
- ▣ Model: A7

Exercise B1: 1/N versus Markowitz on Sectors [Solved]

Task

- Nine sector ETFs, monthly excess returns, 60-month rolling window, OOS January 2004 – July 2026
- Rules: 1/N, MV, MV-LO, GMV; report Sharpe, turnover and tests against 1/N (HAC and bootstrap)

Interpretation

- MV-LO beats 1/N by 0.06 with $p = 0.60$: how long a sample would make this difference significant?

Results ($T = 271$)

	SR	TO	$\hat{\Delta}$	p HAC / boot
1/N	0.66	0.02	–	–
MV	0.26	3.86	–0.40	0.22/0.22
MV-LO	0.72	0.23	+0.06	0.60/0.61
GMV	0.71	0.23	+0.05	0.75/0.76

- MV: volatility 69%, wealth wiped out once
- Bootstrap interval for MV-LO: $[-0.19, 0.28]$
- Standard error 0.12 over 22.6 years; halving it needs four times as many years

Exercise B2: Does Shrinkage Lower GMV Risk? [Solved]

Task

- ▣ Same backtest: GMV with the sample covariance vs GMV-LW (constant-correlation target)
- ▣ GMV aims at low variance: compare OOS volatilities; bootstrap interval for the variance ratio
- ▣ Compare turnover

Interpretation

- ▣ Why is the gain small with 9 assets and 60 months? When would it be larger?

Results

- ▣ OOS volatility: GMV 11.90%, GMV-LW 11.80% (1/N 14.28%)
- ▣ Variance ratio 0.984, 95% interval [0.918, 1.062]: no significant reduction
- ▣ Turnover: 0.234 $\rightarrow$ 0.137 (−41%)
- ▣ $N/T = 0.15$ is small, so S is already decent; the gain grows with N/T and a good target

Exercise B3: ERC and HRP on Sectors [Solved]

Task

- Last 60-month window (to July 2026): ERC and HRP weights; risk shares of 1/N
- OOS Sharpe of ERC and HRP against 1/N, HAC test

Interpretation

- ERC and 1/N have correlation 0.993: is ERC a different strategy or a small tilt?

Results

- 1/N risk shares: XLB 13.8%, XLI 13.4% ... XLP 7.8%
- ERC: XLP 14.8%, XLV 13.9%, XLU 12.5% ... XLB 8.6%
- HRP: XLV 22.6%, XLP 20.3%, XLU 17.4% ... XLB 3.8%
- OOS Sharpe: ERC 0.70 (+0.04, $p = 0.09$), HRP 0.72 (+0.06, $p = 0.19$), 1/N 0.66
- Turnover: 1/N 0.024, ERC 0.028, HRP 0.081

Exercise B4: Shrinkage on 16 ETFs [Proposed]

Solution discussed in class

Task

- ▣ Combined universe (9 sectors + 7 multi-asset ETFs), OOS May 2012 – July 2026
- ▣ Repeat B2: volatilities, variance ratio with bootstrap interval, turnover, mean δ , condition numbers
- ▣ Model: B2

Interpretation

- ▣ Is constant correlation a sensible target for stocks and Treasuries together?

Exercise B5: Risk Parity with Bitcoin [Proposed]

Solution discussed in class

Task

- ▣ Eight multi-asset ETFs with and without Bitcoin, monthly from August 2010, OOS August 2015 – July 2026
- ▣ 1/N, ERC, HRP with and without Bitcoin: Sharpe, volatility, HAC test of the difference
- ▣ Risk share of Bitcoin in 1/N; mean Bitcoin weight in ERC
- ▣ Model: B3; literature: [Petukhina, Trimborn, Härdle & Elendner \(2021\)](#), [Pele et al. \(2023\)](#)

Interpretation

- ▣ Would you expect the past Bitcoin premium to persist? What about survivorship?

Exercise B6: HRP versus GMV on 16 ETFs [Proposed]

Solution discussed in class

Task

- ▣ Combined universe, OOS May 2012 – July 2026
- ▣ Compare HRP and GMV: Sharpe, volatility, turnover, net Sharpe at 50 bp
- ▣ Test H_0 : equal Sharpe ratios (HAC and bootstrap)
- ▣ Model: B1

Interpretation

- ▣ GMV wins on volatility, HRP on Sharpe: which criterion fits which investor?

Exercise B7: Black–Litterman on a Multi-Asset Set [Proposed]

Solution discussed in class

Task

- ▣ Eight multi-asset ETFs, Σ from August 2016 – July 2026; benchmark $1/N$, $\delta_{RA} = 2.5$, $\tau = 0.05$
- ▣ View: GLD minus TLT = 4% a year, Ω as in He & Litterman (2002)
- ▣ Report the implied and posterior view and the new weights
- ▣ Model: A5, A6

Interpretation

- ▣ The weights move far although the view is modest: why?

Exercise B8: The Break-Even Cost [Proposed]

Solution discussed in class

Task

- ▣ Combined universe: MV-LO versus $1/N$
- ▣ Find the cost c (bp per unit of turnover) at which their net Sharpe ratios are equal
- ▣ Relate it to the turnover difference and the gross Sharpe difference
- ▣ Model: B1

Interpretation

- ▣ Retail costs on BVB or US ETFs: is MV-LO worth implementing?

C1: An Optimal Portfolio of BVB Blue Chips in RON [Proposed]

Solution discussed in class

The question

- ▣ A Romanian investor holds BVB blue chips: does diversification **beyond** BET pay?
- ▣ Build 1/N, GMV-LW, ERC and HRP portfolios of eight blue chips, with and without foreign ETFs converted to RON
- ▣ Compare with BET-TR out of sample, with tests

Deliverable

- ▣ A Quantlet folder in the team repository and one slide with the conclusion and its main limitation

The answer should contain

- ▣ Foreign prices converted at the BNR USD/RON rate on common days
- ▣ BET-TR, not BET, as benchmark
- ▣ The RON risk-free rate (or a sensitivity to it)
- ▣ Sharpe tests against BET-TR and against the BVB-only rule
- ▣ Currency risk, costs and taxes

C2: Critique an AI Answer [Proposed]

Solution discussed in class

The prompt sent to an artificial intelligence (AI) assistant

- “prices holds daily adjusted closes of ten sector ETFs. Write Python code for a GMV backtest with monthly rebalancing and give me its annualised Sharpe ratio.”

The assistant's answer

- “Below is an out-of-sample GMV backtest, rebalanced monthly to the GMV weights:”

```
import numpy as np
r = prices.resample('ME').last().pct_change().dropna()
S = r.cov().values
w = np.linalg.solve(S, np.ones(len(S)))
w = w / w.sum()
port = r.dot(w)
sr = port.mean() / port.std() * np.sqrt(252)
print(f"Annualised Sharpe ratio: {sr:.2f}")
```

monthly simple returns
covariance matrix
$\Sigma^{-1} \mathbf{1}$
GMV weights, sum to one
portfolio return

C2: Critique an AI Answer, the Task [Proposed]

Solution discussed in class

- The code runs without an error message, but it contains **three** mistakes typical for backtests
 - ▶ Everything else in the answer is correct
- For each mistake
 - ▶ Name the line and say why it is wrong
 - ▶ Say how you would detect it: a test, a number from A7 or B1, a slide of the lecture
 - ▶ Write the corrected line(s)
- Log each mistake as in `AI_ERRORS.md`
 - ▶ What was asked, what the AI answered, what was wrong, how you found out, the correct version
- Hint: reread the slide “Key Definitions: Evaluation and Inference” and the rule on out-of-sample weights

Common Pitfalls

Design

- ▣ Estimating weights with data from the evaluation month (look-ahead)
- ▣ Reporting in-sample Sharpe ratios of optimised portfolios
- ▣ Ignoring turnover and costs
- ▣ Comparing with BET instead of BET-TR

Statistics

- ▣ Comparing Sharpe ratios without a test
- ▣ i.i.d. tests on dependent returns
- ▣ Many rules and universes, one p -value below 0.05
- ▣ Inverting a near-singular covariance matrix

- ▣ Rule: state the window, the rebalancing frequency, the costs and the benchmark of every backtest

Take-Home Assignment (Team Project)

- ▣ Choose **ten** assets: at least three BVB stocks, two bond ETFs and one commodity or crypto asset
- ▣ Backtest 1/N, GMV-LW, ERC and HRP with a rolling window
 - ▶ Report Sharpe ratio, volatility, maximum drawdown and turnover, gross and net of 20 bp
 - ▶ Test each rule against 1/N (HAC and block bootstrap)
- ▣ Sensitivity: window of 36 versus 60 months
- ▣ Log every AI error you catch in `AI_ERRORS.md` (at least three per project)
- ▣ Deliverables: a GitHub repository with a notebook that runs top-to-bottom and a 2-page summary

Further Study on Quantinar

Advanced courses

- ▣  Hierarchical Clustering
- ▣  Minimum Spanning Tree

Code

- ▣ **Notebook:** `chapter4_seminar_notebook.ipynb` (Colab)
- ▣ **Quantlets:** github.com/danpele/MFM/tree/main/Quantlets/Ch_04

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